System Architecture: Multi-Source Candidate Data Transformer

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1. Ingestion Pipeline & Architecture

The ingestion pipeline is designed around a decoupled, stage-based flow: Ingestion Parsers → Field Normalizers → Schema Validation → Entity Resolution → Conflict Resolution → Configurable Projection. Each module operates as an isolated worker block. System resiliency is handled at the field level; malformed CSV entries, missing fields, or empty PDFs do not crash the pipeline, but trigger null-recovery fallbacks and audit warnings.

2. Canonical Candidate Schema

Rather than storing only the unified scalar values, the Canonical profile encapsulates each attribute in a typed *MergedField[T]* schema. This model explicitly logs metadata for every decision: winning source, raw competing values, normalized values, applied strategy, confidence score, and a detailed reason. This guarantees data provenance and complete system explainability out-of-the-box.

3. Conflict Resolution Engine

Conflict resolution is resolved using a deterministic priority queue:

- **Exact match:** Identical values across all sources (agreement boost).
- **Normalized match:** Match after cleanup (e.g. C++ vs. c plus plus).
- **Source Reliability:** Highest weight source wins (CSV > ATS > Resume > Notes).
- **Completeness:** Longer strings or larger lists win on tie-breaker.
- **Timeline Date:** Chosen based on the latest work experience history end date.
- **Fallback:** Alphabetical fallback + list-union for skills/URLs.

4. Confidence Score Model

Field confidence is calculated using the formula:

$C = \min(R_s * N_s * V_s * A, 1.0)$

Where components represent:

- **R_s (Source Weight):** Loaded from dynamic configuration (default weights: CSV=0.95, ATS=0.90, Resume=0.80, Notes=0.60).
- **N_s (Normalization Success):** 1.0 on success, 0.7 on fallback/failure.
- **V_s (Validation Success):** 1.0 on passed Pydantic rules, 0.5 on failure.
- **A (Agreement Multiplier):** $1.0 + 0.05 * (k - 1)$ for k agreeing sources.

5. Configurable Projection Layer

The canonical record is treated as an immutable ledger. Target representations are generated on-the-fly using a JSON config file. Users can customize field selections, map fields (e.g. first_name → fname), apply post-normalizers (e.g. lowercase emails), choose missing-value policies (fill_null, ignore, default_value), and toggle confidence scores and provenance logs without changing the core engine code.

6. Trade-offs & Edge Cases

- **Trade-off (Complexity vs. Performance):** Running connected-components graph groupings for entity resolution increases CPU time but is necessary to prevent candidate duplicate clusters.
- **Edge Cases (Multiple Employments):** Grouping work experience solely by company risks merging disjoint employments (e.g., worked at Google in 2020, and again in 2024). Our engine checks date-interval overlaps to ensure they are merged only if their timelines intersect; otherwise, they remain separate experiences.